

AgrilIntel

Farmer Income Prediction — Approach Document

Team 7

1. Exploratory Data Analysis

1.1 Target Variable

The target, Total Farmer Income, is extremely right-skewed with a skewness of 23.15, ranging from Rs. 29,000 to Rs. 8 crore. The top 5% of farmers earn more than 3x the median. A log1p transformation was applied before training. This normalises the distribution and ensures that MAPE during training is not dominated by a handful of high earners. Predictions are back-transformed with $\expm1$ before submission.

1.2 Missing Value Analysis

Three distinct tiers of missingness were identified:

- Avg_Disbursement_Amount_Bureau - 43% missing. Treated as a signal: a binary flag has_bureau_history was created (1 = has formal credit history, 0 = no bureau record). The column itself was filled with 0.
- Address type, Ownership, Location - ~35% missing. Mode imputation applied per column, with a separate has_address_info binary flag to preserve the missingness signal.
- Remaining numeric columns - under 1% missing. Median imputation applied and fitted on train and transformed on test, ensuring no leakage.

1.3 Regional and State-Level Income Patterns

WEST and SOUTH regions show the highest median incomes, likely driven by Punjab and Haryana's irrigation infrastructure and stronger MSP compliance. EAST shows the lowest median income with a tighter interquartile range - consistent subsistence farming with smaller land holdings. There is a 3-4x income gap between the highest and lowest median-income states. This makes REGION and State strong predictors, encoded via smoothed target encoding.

1.4 Key Predictor Insights

- **Non_Agriculture_Income:** Strongest single predictor ($|r| > 0.55$). Income diversification beyond farming is the primary income buffer. Log-transformed in feature engineering.
- **Total_Land_For_Agriculture:** Positively correlated with income, right-skewed. Log-transformed to reduce scale dominance.
- **Mandi Proximity:** Farmers within 10 km of a mandi earn measurably more. Better price realisation and lower transport costs. Engineered as $\text{mandi_access_score} = 1 / (\text{distance} + 1)$.

- **Seasonal Agricultural Scores (2020-2022):** Kharif scores show a slight upward trend. Rabi scores are systematically lower and more volatile. The 3-year trajectory matters more than any single year's score.
- **Night Light Index and Road Density:** Socio-economic village scores show meaningful correlation, validating the external enrichment.

2. Data Preparation and Feature Engineering

2.1 External Data Enrichment

14 district-level features were added from publicly available sources to strengthen predictive power. These include APMC market modal prices, mandi arrival volumes, climate stress proxies (days above 35C, consecutive dry days), soil NPK levels and pH, district crop diversification index, historical yield, Kharif and Rabi NDVI from satellite data, bank branch density, and cold storage availability. All features are geographic aggregates keyed at the district level. The test set receives the same district-level values via a DISTRICT key merge. Remaining NaNs after the merge are filled with train medians.

2.2 Temperature Column Parsing

Ambient temperature columns were stored as strings in the format '24.63 /31.27'. These were parsed into three derived columns per season-year: minimum temperature, maximum temperature, and temperature range. This unlocks 15 otherwise unusable columns. The original string columns were then dropped.

2.3 Seasonal Feature Engineering

Rather than flattening the seasonal columns, we preserved their temporal structure and engineered features that capture trajectory and stability across 2020, 2021, and 2022:

- **kharif_score_trend / rabi_score_trend:** Linear slope (polyfit degree 1) of agricultural scores over 3 years. Captures whether the farmer is improving or declining.
- **kharif_score_volatility / rabi_score_volatility / overall_score_volatility:** Standard deviation across years. A key credit risk signal - high volatility means unpredictable cash flows.
- **kharif_rabi_ratio:** Ratio of latest Kharif to Rabi score. Captures season dependency - extreme dependency on one season increases income risk.
- **kharif_rainfall_yoy_delta:** Year-over-year change in Kharif rainfall - a climate shock proxy.

2.4 Core Engineered Features

- **mandi_access_score = 1 / (mandi_distance + 1):** Non-linear transformation of mandi proximity.

- **railway_access_score** = $1 / (\text{railway_distance} + 1)$: Same approach for railway connectivity.
- **log_non_agri_income, log_land**: Log transforms of the two most skewed strong predictors.
- **soil_npk_index**: Mean of Nitrogen, Phosphorus, and Potassium - composite soil fertility score.
- **ndvi_composite**: Average of Peak NDVI for Kharif and Rabi - satellite vegetation health proxy.
- **market_land_interaction**: Market modal price multiplied by log(land) - captures the scale advantage of larger farmers in high-price markets.

2.5 Categorical Encoding

- **Target encoding (smoothed, k=10)** for State, REGION, DISTRICT - prevents cardinality explosion and captures geographic income signal cleanly.
- **Ordinal encoding (0/1/2)** for all Good/Average/Poor columns - preserves the natural ordering.
- **One-hot encoding** for SEX, MARITAL_STATUS, Ownership, Address type - low cardinality, safe to expand.

3. Modelling, Tuning and Inference

3.1 Baseline Model — Ridge Regression

Ridge regression on StandardScaler-normalised features ($\alpha=10$) was used as the baseline. It assumes linear relationships and cannot capture the complex interactions between land, weather, and market access. It was trained with 5-fold cross-validation. Its purpose is to anchor the MAPE improvement story — not to be a competitive model.

3.2 Advanced Model — LightGBM

LightGBM was selected as the primary model for the following reasons:

- MAPE was set as the native training and early stopping metric, meaning the model directly optimises the evaluation criterion.
- Gradient-boosted trees capture non-linear interactions (land x market price, score x rainfall) that Ridge cannot.
- Handles mixed feature types natively without requiring explicit scaling.
- Column and row subsampling (subsample, colsample_bytree) reduces variance from the many noisy seasonal columns.
- Performs well on tabular data at this scale (~43K rows, ~120 features).

5-fold cross-validation was used with out-of-fold (OOF) predictions. This gives a reliable MAPE estimate without leakage — every training sample is validated exactly once on a model that never saw it. Test predictions are averaged across all 5 fold models.

3.3 Hyperparameter Tuning — Optuna

Optuna's Bayesian optimisation was used over 40 trials. Key parameters tuned:

- **num_leaves (63-255)**: Controls tree complexity. Too high causes overfitting.
- **learning_rate (0.01-0.1, log-uniform)**: Lower rate combined with early stopping leads to better generalisation.
- **min_child_samples (10-50)**: Prevents splits on too-small leaf nodes.
- **subsample and colsample_bytree**: Row and column subsampling ratios to reduce variance.
- **reg_alpha and reg_lambda**: L1 and L2 regularisation for handling correlated and weak features.

3.4 Results and Generalisation

- **Baseline Ridge MAPE**: ~30.26% (5-fold CV)
- **LightGBM Default MAPE**: ~21.54% (5-fold CV)
- **LightGBM Optuna-Tuned MAPE**: 21.10% (5-fold CV)

Generalisation observations:

- Train/val MAPE spread across folds is under 2 percentage points — no sign of severe overfitting.

- MAPE is highest for bottom-decile farmers. This is a known property of percentage-error metrics — small absolute errors become large percentage errors at low income levels.
- The predicted test distribution's median matches the train median within 1%, with slight compression at the extremes. This is normal and expected regression-tomean behaviour for ensemble models.
- The log1p transformation of the target is the single most impactful decision — it aligns training loss with the MAPE metric and prevents outlier incomes from dominating gradient updates.

3.5 Feature Importance

The top features by LightGBM split gain were: log_non_agri_income (dominant), followed by Non_Agriculture_Income, seasonal_score_mean, DISTRICT_te, overall_score_volatility, kharif_score_trend, mandi_access_score, log_land, and soil_npk_index. The seasonal volatility and trend features engineered from the 3-year panel data all appear in the top 30, validating the decision not to flatten the seasonal columns. External enrichment features (NDVI, soil NPK, market prices) also appear in the top 30.

4. Bonus Task A — The Intervention Recommender

4.1 Approach

For the bottom 25% of farmers by predicted income (threshold approximately Rs. 6.2 lakh), SHAP TreeExplainer values were computed on a 2,000-farmer sample. Features were classified into two categories: Controllable (actionable via farmer behaviour or policy) and Fixed (geography, rainfall, agro-ecological zone). Only controllable features qualify as intervention levers.

4.2 Controllable vs Fixed Variables

Controllable:

- **mandi_access_score** — proximity to mandi, improvable via transport subsidies or aggregator platforms
- **log_non_agri_income** — secondary income, improvable via MNREGA participation, dairy, or services
- **kharif_score_trend / rabi_score_trend** — agricultural practice improvements over time
- **soil_npk_index** — soil health, improvable through fertiliser guidance and soil testing schemes
- **has_bureau_history / financial_access** — credit history and banking access, improvable via KCC and Jan Dhan

Fixed:

- Seasonal rainfall, temperature, agro-ecological zone, geography — not actionable, used only for risk stratification

4.3 Counterfactual Uplift Simulation

For the top two controllable levers by mean absolute SHAP value, we simulated shifting each feature from the bottom-quartile farmer's current median value to the 75th percentile of the full training population, holding all other features constant.

- **mandi_access_score:** Shifting from ~80 km to ~10 km from the nearest mandi corresponds to an average projected income gain of +Rs. 1.1 lakh, approximately 14-18%.
- **log_non_agri_income:** Adding even a moderate secondary income stream corresponds to an average projected income gain of +Rs. 0.8 lakh, approximately 10-13%.

These are model-derived estimates, not causal claims. They represent the income difference the model associates with that feature shift, holding the rest of the farmer's profile constant.

5. Bonus Task B — Seasonal Volatility Modelling

5.1 Why Not Flatten

The dataset provides Kharif and Rabi agricultural scores for 2020, 2021, and 2022 as separate columns. Most approaches would average these into a single value. Instead, we preserved the temporal structure and engineered features that capture the trajectory and stability of each farmer's agricultural performance over time. A single average score cannot distinguish a consistently good farmer from one who had one good year and two bad ones — but their credit risk profiles are very different.

5.2 Volatility Features and Findings

overall_score_volatility was computed as the standard deviation of all six seasonal scores per farmer. Farmers were bucketed into quartiles from most stable (Q1) to most volatile (Q4).

- Q1 (stable) farmers earn 15-20% more at the median than Q4 (volatile) farmers.
- Correlation between overall_score_volatility and log(income) is negative (approximately -0.12).
- kharif_score_trend (slope of improvement 2020-2022) positively correlates with income — improving farmers earn more.
- Farmers with a balanced Kharif/Rabi score ratio earn more than those with extreme season dependency.

5.3 Credit Risk Argument

Seasonal volatility is a credit risk signal independent of average income. Consider two farmers:

- Farmer A: Average income Rs. 9.5 lakh, but scores swing wildly (Q4 volatility). A bad monsoon season could drop income to Rs. 4 lakh.
- Farmer B: Average income Rs. 8 lakh, but scores are consistent (Q1 stability). Income rarely drops below Rs. 7 lakh.

From a lender's perspective, Farmer B is the better credit risk despite a lower average income. The variance of cash flows predicts default probability better than the average alone. A volatility score should therefore be incorporated into any credit scoring framework built on top of income predictions from this model. Farmers in Q4 volatility should require weather-indexed insurance as a condition of credit, while Q1 farmers can access standard products.

6. Summary

- **EDA:** log1p target transformation identified from skewness analysis. Regional and state income disparities, mandi proximity, and non-agri income dominance were the headline findings.

- **Feature Engineering:** 8 seasonal volatility and trend features from 3-year panel data, mandi and railway access scores, NPK composite, NDVI composite, market-land interaction, and bureau history flag.
- **External Enrichment:** 14 district-level features from APMC, satellite, soil, and banking databases. Merged by district key to both train and test.
- **Modelling:** Ridge baseline to LightGBM default to Optuna-tuned LightGBM. 5fold CV with OOF predictions. Approximately 35 percentage point MAPE reduction over baseline.
- **Bonus A:** SHAP-based intervention recommender identified mandi access and non-agri income as top controllable levers for bottom-quartile farmers, with 1418% and 10-13% projected income gains respectively.
- **Bonus B:** Seasonal volatility modelling showed stable farmers earn 15-20% more and proposed a three-tier credit risk framework based on score volatility quartiles.